Statement of Purpose

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Research interests. I am applying to pursue a PhD in [Program name]at [University name], where I intend to study questions at the intersection of public, labour and health economics. My specific research agenda asks how large-scale government programs can better achieve their desired policy objectives, and the externalities created by such programs in the pursuit of these goals. So far, I have begun to tackle this research agenda in three key ways. Firstly, in my undergraduate Honours thesis that was published as a central bank working paper, I evaluated the effects of unconventional monetary policy – which constitutes large central bank investment in instilling market confidence – on financial stability in an emerging market context. Secondly, through my pre-doctoral research I am examining how the introduction of Australia's National Disability Insurance Scheme (NDIS) changed the labour market composition and returns to disability work, and potential effects on care. Finally, in my independent research, I have been exploring the efficacy of large-scale supply-side education policies on improving individual employment outcomes as well as labour market shortages. Methodologically, I seek to use microdata, particularly Australian microdata which I have used as part of my pre-doctoral research, with theoretical models to imbue greater intuition to empirical findings.

Pre-doctoral and professional experience. I have been broadening my research toolkit working for A/Prof Manasi Deshpande as a pre-doctoral researcher, where we have been exploiting the staggered rollout of NDIS to examine how this large-scale government program funding disability services causally changed labour supply in disability work, wages, related professions such as nursing and therapy, and categorise the returns to NDIS among workers, firms and households. Through this research I have gained significant experience working with large administrative datasets, applying an array of causal inference methods, as well as implementing a LASSO model to identify a clean 'risk set' of individuals for whom NDIS may have induced labour supply. During my pre-doctoral position, I audited A/Prof Deshpande's PhD elective course on Empirical Topics in Social Insurance, where I learned about the challenges in policy design, targeting and heterogeneity in outcomes, but also about innovative techniques to generate variation, assign treatment, and study these programs in ways that meaningfully contribute to the literature. Overall, my pre-doctoral experience has taught me how to think deeply, creatively and methodically about complex economics questions, and to conduct economic research deliberately and credibly. Prior to my pre-doctoral position, I worked at the Reserve Bank of Australia (RBA), Australia's equivalent of the Federal Reserve. In this role, I contributed to and led an array of analytical research, most recently applying contingent valuation and textual analysis in a large-scale population representative survey, which is in the process of becoming a central bank working paper.

Future research. I have recently taken an interest in the effect of 'supply-side' education policies on labour market matching, and I believe a policy initiative from Australia offers a valuable setting to study it. Since 2019, the state of Victoria has offered 'free TAFE', a program where residents can enrol at no cost in a designated list of vocational courses in areas deemed to have labour shortages. Examples include disability care, nursing, construction, and early childhood care.

I believe examining free TAFE could yield at least two useful contributions to the literature. In the first instance, it could yield insights into whether preferentially subsidised training policies are a legitimate policy lever to upskill and reskill individuals in growth sectors, especially when

implemented at scale, and if there are general equilibrium effects of this policy. For example, I am interested in if wage premia in these sectors changed in response to the increased supply of skilled workers, or if this creates other shifts in the labour market composition. I intend to answer this component of the research by exploiting the unanticipated announcement of the program in Victoria and the staggered addition and removal of eligible courses. I intend to compare at least the following outcomes relative to the counterfactual in other Australian states: conversions of enrolments to completions between individuals who enrolled during free-TAFE across states, likelihood of being employed within twelve months of course completion (and separately, likelihood of being employed in a related occupation to the completed course), difference in wages between free-TAFE completers and non free-TAFE completers over time.

Secondly, while most supply-side education and training programs restrict eligibility to a specific cohort, such as the unemployed, or workers needing particular skills, free TAFE pairs near-universal participant eligibility with a narrow, government-selected set of eligible courses. This design to my knowledge has not previously been investigated in the education literature, particularly in a quasi-experimental setting. As such, I believe it could be highly valuable to explore the extent to which 'self-targeting' can be leveraged effectively. I have been attempting this by using the full sample of vocational course enrollees to identify differential enrolment trends among the following groups: self-reported labour force status at enrolment (e.g., employed, unemployed seeking work, unemployed not seeking work), age group, gender, and highest educational level. This then motivates identifying a suitable 'risk set' of individuals who are ex-ante likely target enrollees from the government's perspective (e.g., young people on unemployment benefits), and comparing the existence of, and magnitude of, the first stage on enrolments on all working-age Australians to this risk set.

I have been using Australian administrative data from the Australian Bureau of Statistics (ABS) that link unit-level records, which allows me to observe individuals' prior education history, socio-economic background, parental income, health, and employment outcomes over time.